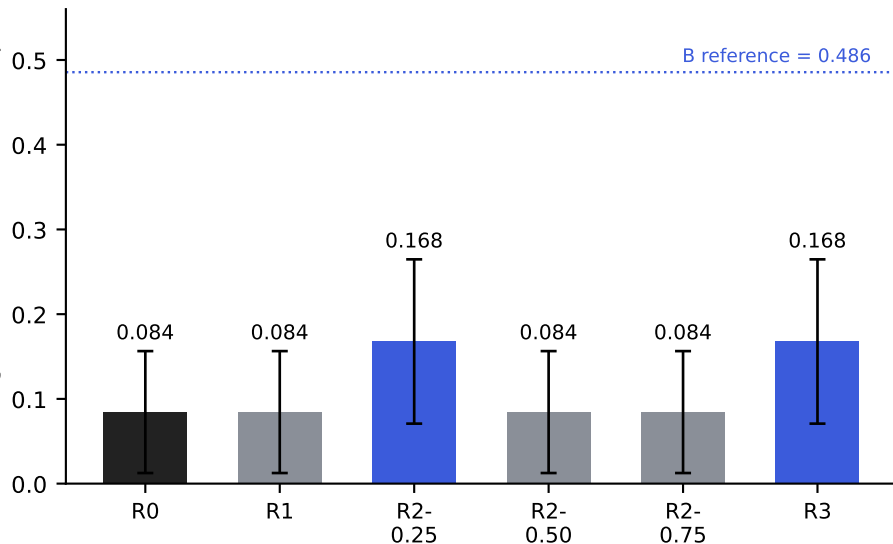


Figure 4. (a) R3 raises the synthetic metric to 0.168; R3 – R0 mean shift +0.083 (95% CI [0.011, 0.155], $p = 0.0253$), on only 5 of 60 non-zero pairs; R3 is NOT PROMOTED, production stays R0. Blue bars = pre-registered verdict “promising”, grey = “no satisfactory calibration”. (b) The primary negative result is not driven by the two revenue-proxy scenarios (excluding them $p \approx 2.2 \times 10^{-7}$; a re-analysis of frozen observations, not a new experiment); bars labelled wins / ties / losses. Synthetic; no real-world validation.

(a) Pre-registered risk-heuristic recalibration (experiment b8516eef)



(b) Robustness to the KPI-proxy scenarios

